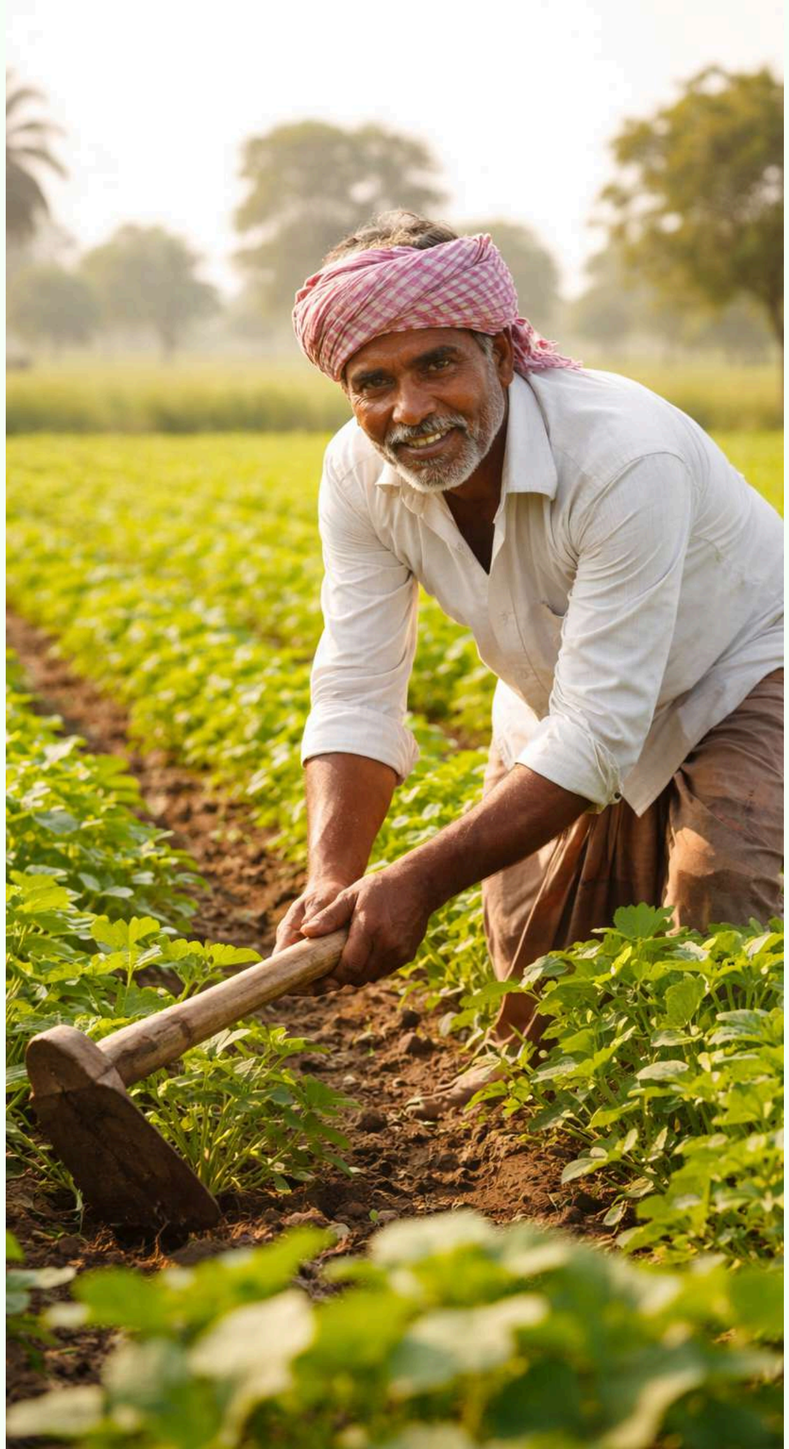
Krishi Mitra

AI-Based Farmer Query Support & Advisory System

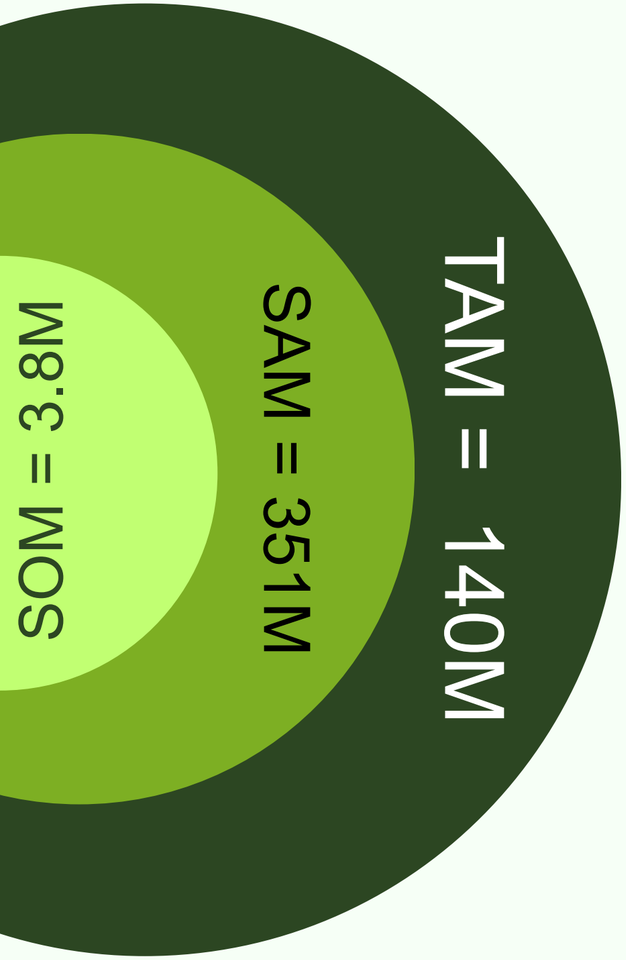


KRISHI MITRA

AI-BASED FARMER QUERY SUPPORT AND ADVISORY SYSTEM

TEAM NAME : NEXWINNER

Market Landscape & Core Hurdles



Market Analysis

TAM (Total Addressable Market): India's 140 million farmers. The broader Indian agritech market is projected to reach $38.56 billion by 2026.

SAM (Serviceable Addressable Market): The 351 million+ rural internet users in India (growing at 15% YoY). Focus on those using smartphones for fintech and e-learning.

SOM (Serviceable Obtainable Market): Targeting 1.8 to 3.88 million farmers initially—aligned with current AI-enabled agricultural network reach in India.

Problem Understanding



The Guidance Gap

15-25% of crop losses are due to a lack of timely expert guidance, not just climate.



Accessibility Crisis

70% of Indian farmers lack access to AI tools or professional advisory.



Economic Impact

Suboptimal decision-making leads to an annual economic loss of over ₹1.25 lakh crore.

Pain Point Prioritization



RAGHAV

The Digitally-Excluded Traditionalist | **Land** 2Acres | **Age** 32

"Mobile mein sab English mein hota hai, kuch samajh nahi aata. Agar koi bol kar bata de toh kaam aasaan ho jaye"



SAKSHI

The High-Risk Cultivator | **Land** 5Acres | **Age** 32

"Mausam ka bharosa nahi. Agar 12 ghante pehle pata chal jaye ki barish hogi, toh main ₹1000 ka nuksan bacha sakti hoon."

Govt. Scheme Awareness → **Input Cost Inflation** → **Market Price Opacity**

1. While critical for growth, the immediate survival threat is lower than crop loss.
2. Biased retailer recommendations increase the financial burden on farmers.
3. Results in distress sales at mandis due to lack of bargaining power.

Language Barrier → **Crop Disease Misdiagnosis** → **Localized Weather Alerts**

4. Creates a fundamental exclusion from existing digital tools.
5. Leads to incorrect pesticide use, financial loss, and reduced trust in tech.
6. Can lead to 100% crop loss without 6–24 hour advance warning.

From Foundation to Future: A Staged Roadmap for Impact and Adoption

Feature Modules Prioritized by Immediate Impact and Long-term Growth



AI Disease Detection

Technology Stack

CNN (Convolutional Neural Networks)

Core User Benefit

Provides **unbiased, affordable,** and **instant crop diagnosis.** (Prevents 15-25% crop loss)

Strategic Urgency

CRITICAL

(Foundation for rapid tech adoption)



Multilingual Voice AI

HIGH (Foundation for deep user engagement)

Technology Stack

NLP Models | Speech-to-Text

Provides unbiased, affordable, and **instant crop diagnosis.** (Prevents 15-25% crop loss)



Severity Alert System

Triggers immediate **expert calls** for high-risk Red Alerts.

Logic-based AI Agents

HIGH

(Immediate intervention)

Economic Weather Layer

Hyperlocal APIs | Financial Modeling

Translates weather data into specific **'Loss/Gain' rupee values.**

MEDIUM (Improves daily efficiency)

Market Intelligence

Real-time Data Scraping | API Integration

Increases **bargaining power... reduces distress selling.**

MEDIUM (Price compare across mandis)

Govt. Scheme Engine

AI-Driven Filtering | Profile Recommendation

Matches farmer profiles to **eligible subsidies.**

LOW/MODERATE (Long-term growth)

Competitor Analysis

Feature	Plantix	DeHaat	Cropin	Krishi Mitra
Disease detection	✓	✗	✓	✓
Personalized Advice	✗	−	−	✓
Vocal Support (Local Language)	✗	✗	✗	✓
Predictive AI	✗	✗	−	✓
Offline Mode	✗	✗	✗	✓
End to End Solution	✗	✓	−	✓
Small Farmer accessibility	✓	✓	✗	✓

✓ Supported

− Partial/ Limited

✗ Not Supported

The Accessibility Gap: Highlight that major players like CropIn and Plantix lack Voice Support and Offline Mode, which are critical for our rural personas.

The Prediction Gap: Note that while others do diagnosis, Krishi-Mitra is the only one offering Predictive AI for future risks.

Score Summary: Mention Krishi-Mitra's 7/7 score compared to the industry average of ~3/7.

Competitive Strategy & Business Model

Market Positioning: Competitor Comparison

Competitor	Their Focus Area	Krishi-Mitra's Competitive Advantage
CropIn	Enterprise/B2B SaaS	Focus on Direct, On-ground Farmer Assistance
DeHaat	Commerce-driven (Selling Seeds/Pesticides)	Provides Unbiased, Advisory-first AI Recommendations
Plantix	Diagnosis-focused (Disease Detection)	Offers a Complete, Integrated Ecosystem

Our Unique Value Proposition (UVP)

- **Farmer-Centric AI:** Moves beyond tech to practical, voice-enabled guidance.
- **Government Integration:** Bridges the last-mile gap for national welfare schemes.
- **Ecosystem-in-a-Pocket:** Unified platform with advisory, market intelligence, and community collaboration.

Monetization: Sustainable Growth Engine

1. B2B Partnerships

Primary revenue through collaborations for context-aware product placement.

2. Freemium Model

Core AI remains Free; Premium analytics via manageable subscriptions.

3. Government Tie-Ups

Leveraging digitization programs to build trust and accelerate B2C adoption.

4. Data Intelligence

Providing anonymized regional insights to research and institutional stakeholders.

‘Go-To-Market Strategy: Phased Rollout to Capture Our 3.8M SOM’



Phase 1: Hyperlocal Pilot (Months 0–6)

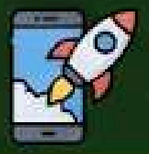
Focus: *Building Trust & On-Ground Validation*

Target Region: Bihar (Wheat) & Kerala (Cash Crops)

Acquisition Channels:

- **FPO & Panchayat Tie-ups:** Onboarding entire farming clusters at once.
- **“Digital Mitra” Program:** Hiring local youth/agri-students to physically demonstrate the app in villages.

Core Metric: 50,000 Active Users | 80% Disease Detection Accuracy validation



Phase 2: Digital Expansion (Months 6–12)

Focus: *Viral Mechanics & Freemium Adoption*

Target Region: Expansion into UP, MP, and Maharashtra.

Acquisition Channels:

- **WhatsApp Voice-Bot:** Allowing farmers to send voice notes/images directly via WhatsApp for frictionless onboarding.
- **Vernacular Social Media:** YouTube Shorts and Facebook ads in Hindi/Malayalam featuring local farmer success stories.
- **Referral Engine:** Free premium weather/market insights for inviting 3 neighbors.

Core Metric: 500,000 Active Users | 20% Premium Conversion



Phase 3: Ecosystem Scale (Months 12–18+)

Focus: *B2B Integration & Profitability*

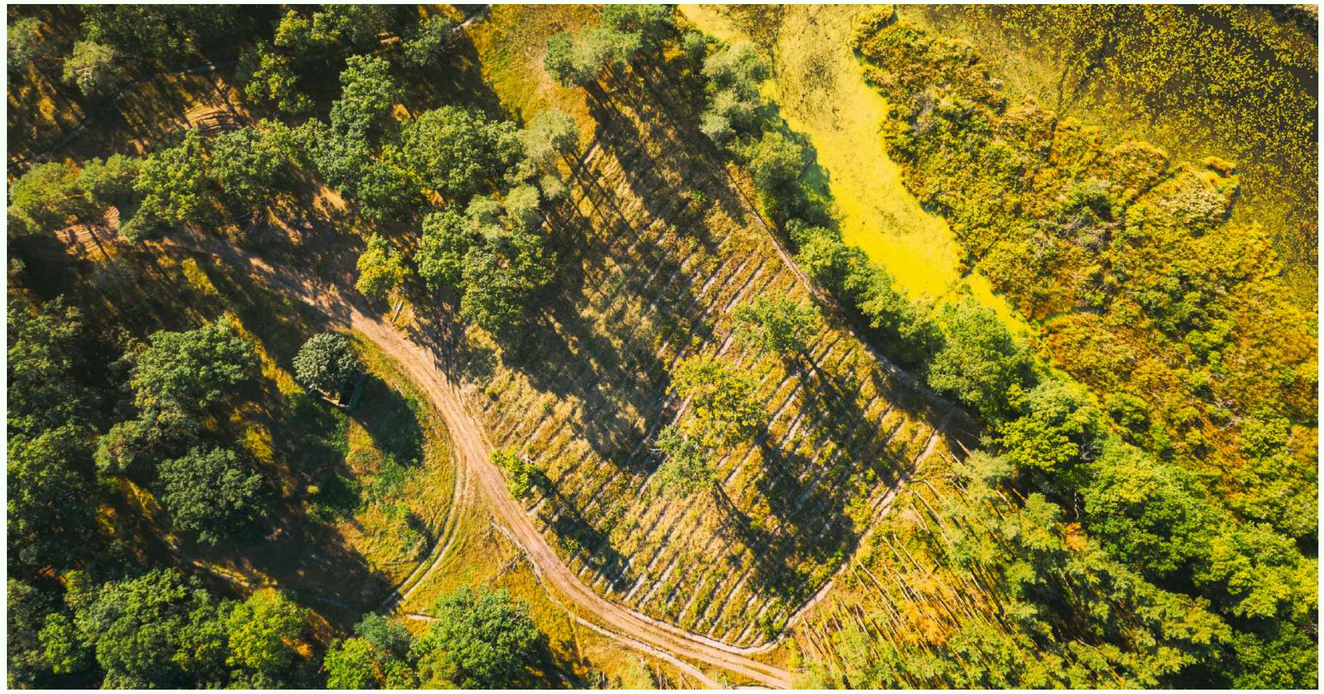
Target Region: Pan-India deployment across 10+ regional languages.

Acquisition Channels:

- **Agri-Input B2B Partnerships:** Co-branding with seed/fertilizer companies (e.g., printed QR codes on fertilizer bags to download the app).
- **Government KVK Integration:** Partnering with Krishi Vigyan Kendras to use the platform as their official advisory distribution tool.

Core Metric: 3.8M Active Users (Capturing full SOM) | Break-even MRR

The Trust Foundation | *Strategic Enablers:* 🤝 Local NGOs | 🏛️ Agritech Incubators | 📱 Telecom Bundling (e.g., Jio/Airtel rural data packs)



THANK YOU

Reference



New Plant Diseases Dataset

Image dataset containing different healthy and unhealthy crop leaves.

kaggle.com

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